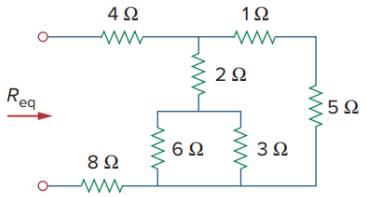
1

Example: Find R_{eq} for the circuit shown in Fig.



Solution:

The 6- Ω and 3-Ω resistors are in parallel, so their equivalent resistance is

$$6\ \Omega \parallel 3\Omega = \frac{6 \times 3}{6+3} = 2\ \Omega$$

The 1-Ω and 5-Ω resistors are in series;

$$1\ \Omega + 5\ \Omega = 6\ \Omega$$

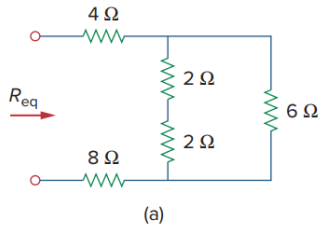
The two 2-Ω resistors are in series,

$$2\ \Omega + 2\ \Omega = 4\ \Omega$$

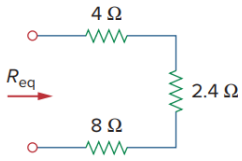
This 4-Ω resistor is now in parallel with the 6-Ω resistor in Fig. (a);

$$4\ \Omega \parallel 6\ \Omega = \frac{4 \times 6}{4+6} = 2.4\ \Omega$$

$$\boxed{R_{eq} = 4\ \Omega + 2.4\ \Omega + 8\ \Omega = 14.4\ \Omega}$$



(a)



(b)

2

- ***Example:*** Determine the value of voltage V shown in the figure.

$I = V/R_{eq}$

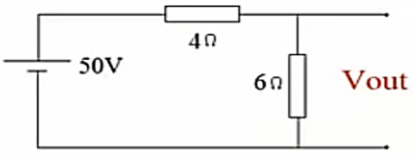
$V = 50$ V

$R_{eq} = 4 + 6 = 10\Omega$

$I = 50/10 = 5$ A

$V_{out} = 5 \times 6 = 30$ volt

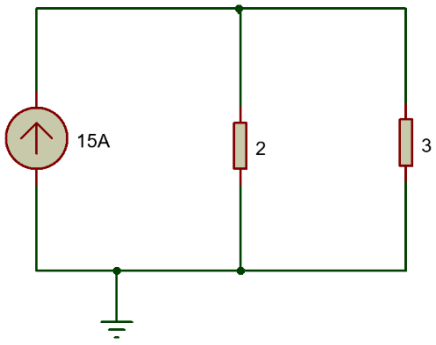
Or;



$$V_{out} = 50\left(\frac{6}{4+6}\right) = 30\ V$$

3

Determine the values of i_1, *and* i_2



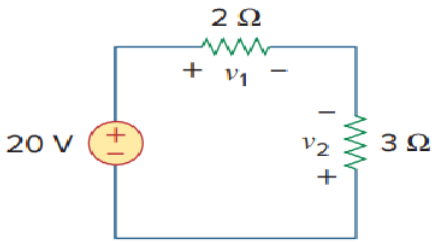
- Applying Current divider law:

$$i_1 = 15\frac{3}{2+3} = 9A$$

$$i_2 = 15\frac{2}{2+3} = 6A$$

4

Example: For the circuit in Fig. (a), find voltages v_1 and v_2.



(a)

Solution:

To find v_1 and v_2 we apply Ohm's law and Kirchhoff's voltage law.
Assume that current i flows through the loop as shown in Fig. (b).
From Ohm's law,

$$v_1 = 2i, \qquad v_2 = -3i$$

Applying KVL around the loop gives

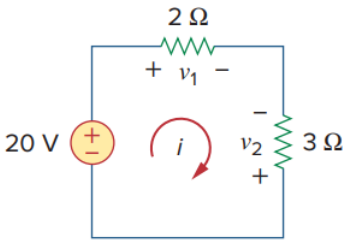
$$-20 + v_1 - v_2 = 0$$

Substituting.

$$-20 + 2i + 3i = 0 \quad \text{or} \quad 5i = 20 \quad \Rightarrow \quad i = 4\ \text{A}$$

Substituting i in Eq. finally gives

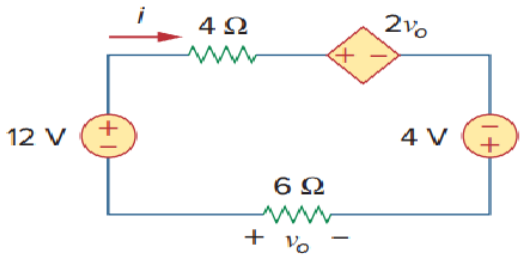
$$v_1 = 8\ \text{V}, \qquad v_2 = -12\ \text{V}$$



(b)

5

Example: Determine v_o and i in the circuit shown in Fig.



(a)

Solution:

We apply KVL around the loop as shown in Fig. (b). The result is

$$-12 + 4i + 2v_o - 4 + 6i = 0$$

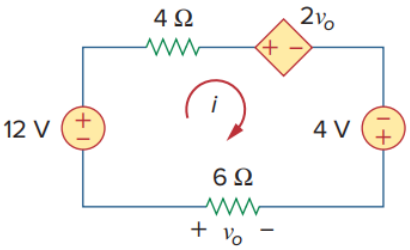
Applying Ohm's law to the 6-Ω resistor gives

$$v_o = -6i$$

Substituting:

$$-16 + 10i - 12i = 0 \quad \Rightarrow \quad i = -8\ \text{A}$$

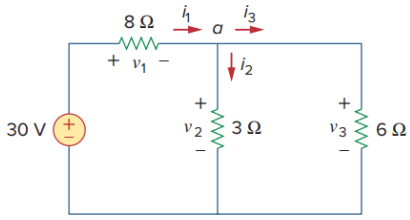
and $v_o = 48$ V.



(b)

6

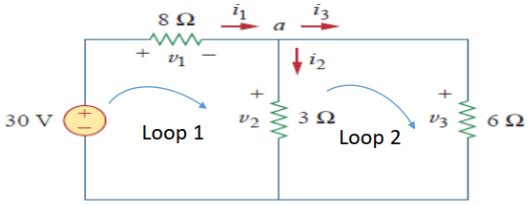
Example: Find currents and voltages in the circuit shown in Fig.



Solution:

Applying ohm's law:

$V_1 = 8\ i_1$
$V_2 = 3\ i_2$
$V_3 = 6\ i_3$



Applying KCL at node a:

$i_1 - i_2 - i_3 = 0$ ⇨ eqn (1)

Applying KVL at loop 1:

$30 = 8\ i_1 + 3\ i_2$

$i_1 = \frac{30 - 3i_2}{8}$ ⇨ eqn (2)

Applying KVL at loop 2:

$-3\ i_2 + 6\ i_3 = 0$

$i_3 = \frac{i_2}{2}$ ⇨ eqn (3)

By Substituting from eq (1) & eq(2) in eq (3) :

$\rightarrow \frac{30 - 3i_2}{8} - i_2 - \frac{i_2}{2} = 0$

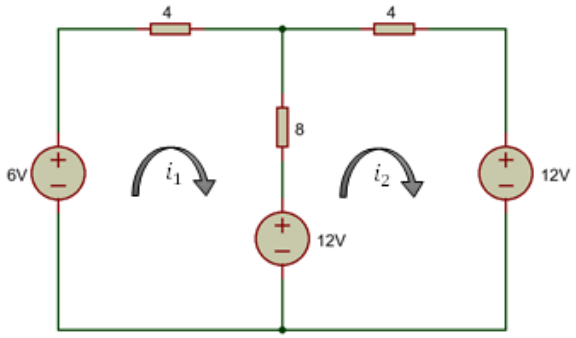
$i_2 = 2\ A$

$\therefore i_1 = i_1 = \frac{30 - 3i_2}{8} = 3\ A$

$\therefore i_3 = i_3 = \frac{i_2}{2} = 1\ A$

$V_1 = 8\ i_1 = 8*3 = 24\ V$
$V_2 = 3\ i_2 = 3*2 = 6\ V$
$V_3 = 6\ i_3 = 6*1 = 6\ V$

7 Find the current loops i_1, and i_2 for the next circuit:



- Applying KVL on loop 1: $\sum V = 0$

$$4i_1 + 8(i_1 - i_2) + 12 - 6 = 0$$
$$12i_1 - 8i_2 = -6 \rightarrow (eq.1)$$

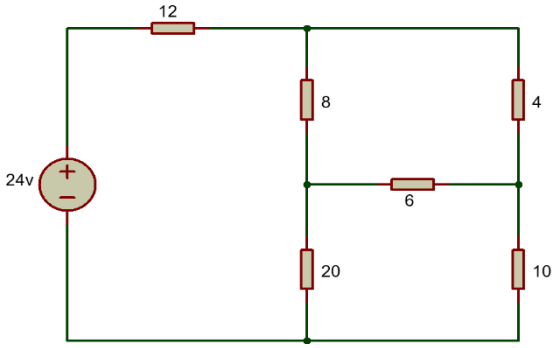
- Applying KVL on loop 2: $\sum V = 0$

$$4i_2 + 8(i_2 - i_1) + 12 - 12 = 0$$
$$-8i_1 + 12i_2 = 0 \rightarrow (eq.2)$$

- Solving the two equations:

$$i_1 = -0.9A = -900mA$$
$$i_2 = -0.6A = -600mA$$

8 Find the current loops i_1, i_2, and i_3 for the next circuit:



- Applying KVL on loop 1: $\sum V = 0$

$$12i_1 + 8(i_1 - i_2) + 20(i_1 - i_3) - 24 = 0$$
$$40i_1 - 8i_2 - 20i_3 = 24 \rightarrow (eq.1)$$

- Applying KVL on loop 2: $\sum V = 0$

$$4i_2 + 8(i_2 - i_1) + 6(i_2 - i_3) = 0$$
$$-8i_1 + 18i_2 - 6i_3 = 0 \rightarrow (eq.2)$$

- Applying KVL on loop 3: $\sum V = 0$

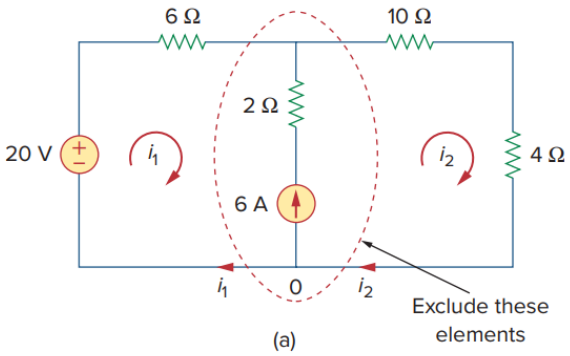
$$10i_3 + 20(i_3 - i_1) + 6(i_3 - i_2) = 0$$
$$-20i_1 - 6i_2 + 36i_3 = 0 \rightarrow (eq.3)$$

- Solving the three equations:

$$i_1 = 1.123A\,,\ i_2 = 0.75A = 750mA\,,\ i_3 = 0.75A = 750mA$$

9

Example: Find i_1 ,i_2 in the circuit :



(a)

Solution:

applying KVL to the supermesh in Fig. (b)

$$-20 + 6i_1 + 10i_2 + 4i_2 = 0$$

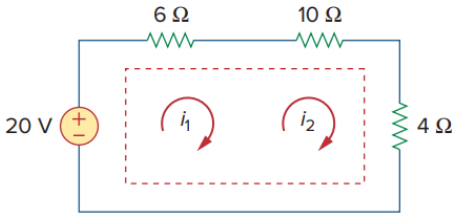
$$6i_1 + 14i_2 = 20$$

Applying KCL to node 0 in Fig. (a)

$$i_1 - i_2 = -6 \quad \Rightarrow \quad i_2 = i_1 + 6$$

Solving Eqs.

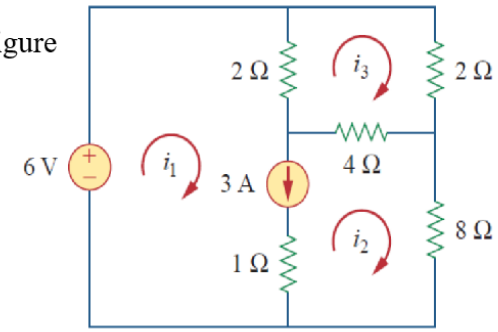
$$i_1 = -3.2\ \text{A}, \qquad i_2 = 2.8\ \text{A}$$



(b)

10

Problem: Use mesh analysis to determine i_1, i_2, and i_3 in figure



Solution:

From supermesh:

$$i_1 - i_2 = 3A$$

Using mesh analysis:

Mesh 1

$$6 = 2(i_1 - i_3) + 4(i_2 - i_3) + 8i_2$$

$$2i_1 + 12\,i_2 - 6i_3 = 6$$

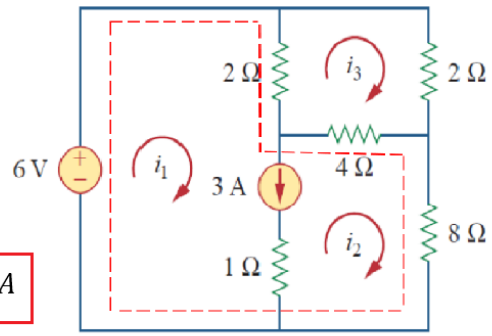
Mesh 2

$$2(i_3 - i_1) + 4(i_3 - i_2) + 2i_3 = 0$$

$$8i_3 - 2i_1 - 4i_3 = 0$$

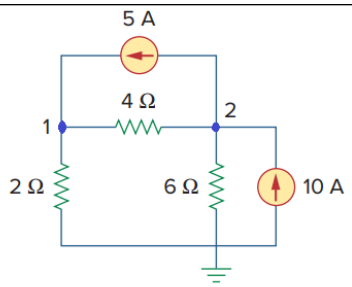
Solve the equations:

$$i_1 = 3.47A \qquad i_2 = 0.473\ A \qquad i_3 = 1.105A$$



11

Example: Calculate the node voltages in the circuit shown in Fig.



(a)

Solution: At node 1, applying KCL and Ohm's law gives

$$i_1 = i_2 + i_3 \quad \Rightarrow \quad 5 = \frac{v_1 - v_2}{4} + \frac{v_1 - 0}{2}$$

Multiplying each term in the last equation by 4, we obtain

$$20 = v_1 - v_2 + 2v_1 \quad \Rightarrow \quad \boxed{3v_1 - v_2 = 20} \quad \Rightarrow (1)$$

At node 2,

$$i_2 + i_4 = i_1 + i_5 \quad \Rightarrow \quad \frac{v_1 - v_2}{4} + 10 = 5 + \frac{v_2 - 0}{6}$$

$$3v_1 - 3v_2 + 120 = 60 + 2v_2 \quad \Rightarrow \quad \boxed{-3v_1 + 5v_2 = 60} \quad \Rightarrow (2)$$

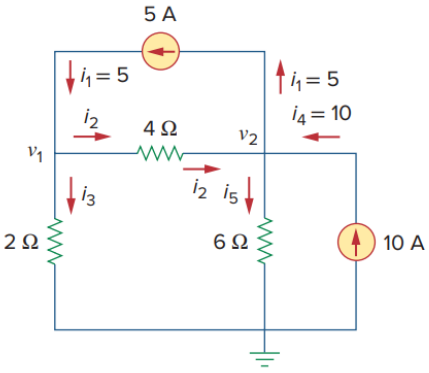
obtain the values of v_1 and v_2.

Method 1: Solve Eq(1) & Eq(2)

$$4v_2 = 80 \quad \Rightarrow \quad \boxed{v_2 = 20 \text{ V}}$$

Substituting $v_2 = 20$ in Eq. (1) gives

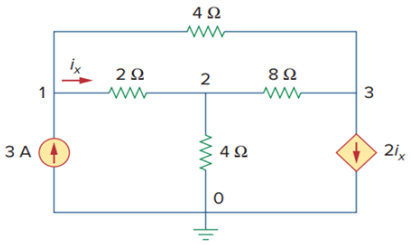
$$3v_1 - 20 = 20 \quad \Rightarrow \quad \boxed{v_1 = \frac{40}{3} = 13.333 \text{ V}}$$



(b)

12

Example: Determine the voltages at the nodes in Fig.



(a)

Solution:

At node 1, $3 = i_1 + i_x \quad \Rightarrow \quad 3 = \frac{v_1 - v_3}{4} + \frac{v_1 - v_2}{2}$

$$\boxed{3v_1 - 2v_2 - v_3 = 12} \quad \Rightarrow (1)$$

At node 2,

$$i_x = i_2 + i_3 \quad \Rightarrow \quad \frac{v_1 - v_2}{2} = \frac{v_2 - v_3}{8} + \frac{v_2 - 0}{4}$$

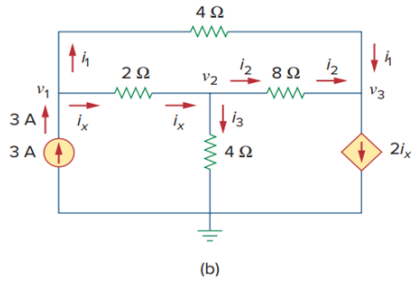
$$\boxed{-4v_1 + 7v_2 - v_3 = 0} \quad \Rightarrow (2)$$

At node 3,

$$i_1 + i_2 = 2i_x \quad \Rightarrow \quad \frac{v_1 - v_3}{4} + \frac{v_2 - v_3}{8} = \frac{2(v_1 - v_2)}{2}$$

$$\boxed{2v_1 - 3v_2 + v_3 = 0} \quad \Rightarrow (3)$$

solve the three equations to solve to get the node voltages v_1, v_2, and v_3.



(b)

Method 1

Using the elimination technique, we add Eqs (1) and (2).

$$5v_1 - 5v_2 = 12 \quad \Rightarrow \quad v_1 - v_2 = \frac{12}{5} = 2.4 \quad \Rightarrow (4)$$

Adding Eqs. (2) and (3) gives

$$-2v_1 + 4v_2 = 0 \quad \Rightarrow \quad v_1 = 2v_2 \quad \Rightarrow (5)$$

Substituting Eq. (5) into Eq. (4) yields

$$2v_2 - v_2 = 2.4 \quad \Rightarrow \quad v_2 = 2.4, \quad v_1 = 2v_2 = 4.8 \text{ V}$$

From Eq. (3), we get

$$v_3 = 3v_2 - 2v_1 = 3v_2 - 4v_2 = -v_2 = -2.4 \text{ V}$$

Thus,

$$v_1 = 4.8 \text{ V}, \quad v_2 = 2.4 \text{ V}, \quad v_3 = -2.4 \text{ V}$$